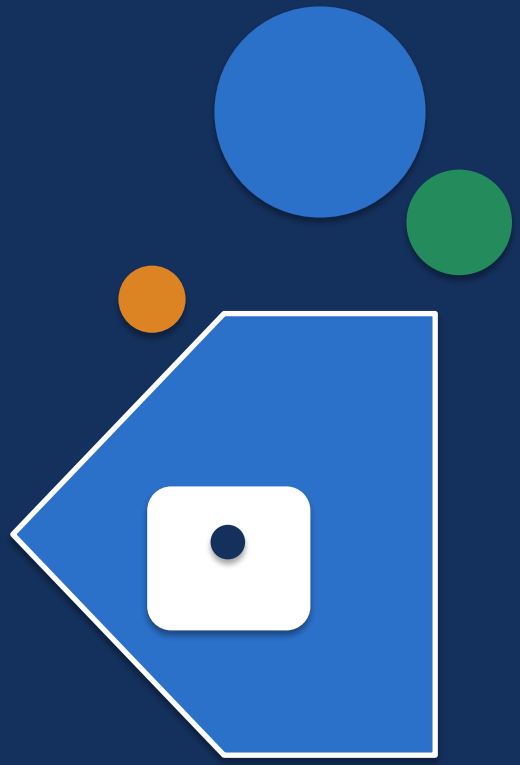
CLOUD SECURITY FUNDAMENTALS

Protecting Data and Applications



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Course Code : CSA1515

1. What Is Cloud Security?

Definition and important security tools

Cloud security protects online data, applications, and systems.

Authentication checks who the user is.

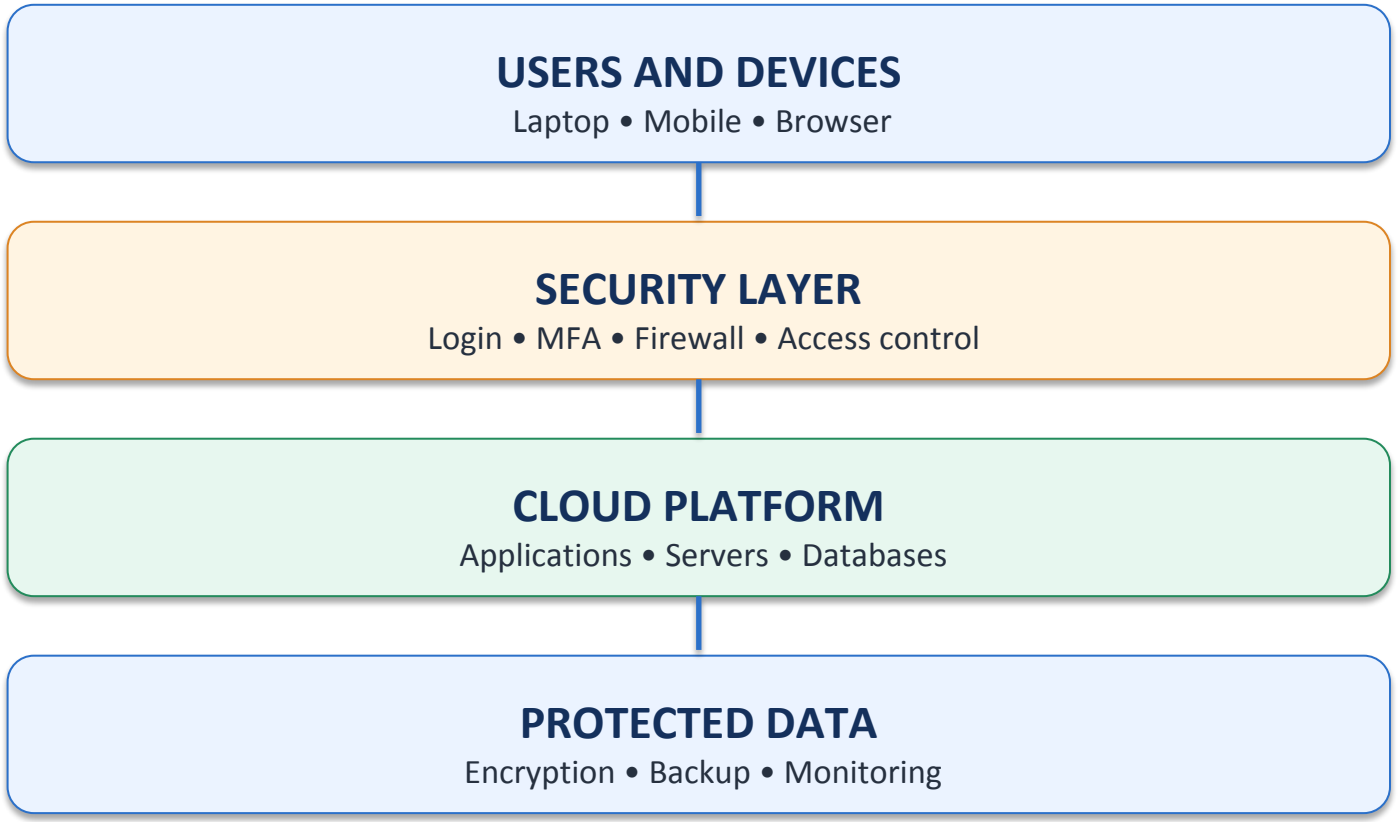
Authorization decides what the user can access.

Encryption protects data by making it unreadable to outsiders.

Backups help restore files if they are lost.

2. Cloud Security Architecture

Security layers work together to protect cloud data



3. How Does It Work?

A simple five-step process

1

1. Login

User enters login details.

2

2. Verify

System checks identity.

3

3. Check access

Permissions are checked.

4

4. Use safely

Authorized data is accessed.

5

5. Monitor

Activities are checked; backups support recovery.

4. Why Is It Important?

Cloud security is used across many industries

HEALTHCARE
Protects patient records

BANKING
Protects accounts and payments

EDUCATION
Secures student files and results

E-COMMERCE
Protects customer details and orders

5. Practical Example: College Files

Scenario: a college stores assignments and mark sheets in cloud storage

Give access only to authorized students and teachers.

Use strong passwords and multi-factor authentication.

Encrypt files to help protect private information.

Keep backups and monitor file access.

Result: student files are better protected from unwanted access and loss.

6. Advantages, Challenges & Conclusion

ADVANTAGES

- Protects sensitive data
- Helps detect threats
- Supports backup and recovery

CHALLENGES

- Phishing and stolen passwords
- Incorrect access settings
- Privacy and compliance risks

CONCLUSION

Strong login security, access control, encryption, backups, and monitoring help protect cloud systems.

THANK YOU